

Exact Dynamics of a Three-Dimensional Keller Map

Degree growth, Newton faces, shear spectra,
and arithmetic escape

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The first dynamical degree of the repository's explicit map is

$$\lambda_1 = 3 + \sqrt{10}.$$

Its iterate degrees satisfy an exact second-order recurrence. Polynomial source shears give an unbounded, explicitly computed spectrum while preserving the generic degree three and the Keller condition.

Status. Complete mathematical arguments and exact companion checks are supplied. The new results have not been peer reviewed or formalized in Lean or Rocq. The original map, its determinant, its collisions, and the source-shear construction are credited to prior work. An exhaustive priority claim is not made.

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[write] Written into ProveIt's research-report collection (category `jacobian-conjecture`) from batch 70, manuscript 06, a single manuscript; provenance and editorial changes are in Appendix C. None of its new results is formalized.

Abstract

We study the iterates of the explicit three-dimensional polynomial map in ProveIt’s Jacobian-conjecture development. We prove that their degrees are $1, 7, 43, 265, \dots$, satisfy $d_{n+2} = 6d_{n+1} + d_n$, and have first dynamical degree $3 + \sqrt{10}$. This establishes the value suggested in Liam Giannini’s publicly available July 2026 record description; it is not a claim about the contents of an inaccessible full manuscript or about global priority. The proof gives considerably more than a limit: all positive weighted degrees are explicit, the upper Newton geometry has just two vertices per coordinate, and every initial form on the intervening wall is a monomial times a power of $xz + 3y$.

For every nonzero polynomial h of degree m , the source shear $T_h(x, y, z) = (x, y, z + y^2 h(xy))$ produces a map $F_h = F \circ T_h$ with

$$\lambda_1(F_h) = \frac{2m + 7 + \sqrt{4m^2 + 32m + 61}}{2}.$$

Thus one tame left–right equivalence class contains Keller maps of fixed generic degree three and unbounded first dynamical degree. We also determine the different degree laws in characteristic three. Finally, a monomial-dominance theorem constructs a forward-invariant complex escape region with a precise logarithmic asymptotic. For integral coefficients it yields a Zariski-dense set of integral initial points whose arithmetic degree equals the first dynamical degree. The package contains thirteen groups of exact checks, a source and claim ledger, a formalization roadmap, and ten further research questions.

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1 The question, the baseline, and the contributions

1.1 A concrete dynamical question

A polynomial map can have three quite different kinds of degree. Its *ordinary degree* is the largest total degree of a component. Its *generic degree* is the degree of the induced extension of rational function fields; in characteristic zero it counts the points of a generic fiber. Its *first dynamical degree* measures exponential growth of ordinary degrees under iteration. The present article separates these quantities and computes the third one exactly for a specific family whose first two are already understood.

Throughout the characteristic-zero parts, $\mathbb{K}$ is a field of characteristic zero. Put

$$u = 1 + xy, \quad H = u^2z + y^2(1 + 3u), \quad (1.1)$$

$$F(x, y, z) = (P, Q, R) = (uH, y + 3xH, x(5 - 3u - x^2z)). \quad (1.2)$$

This is precisely the map used in the repository's Algebra/JacobianConjecture project, not a rescaled or permuted version. The repository supplies Lean and Rocq proofs of its constant Jacobian determinant -2 and an explicit collision [1]. Gao's Theorem 3.3 discusses the same map and its generic degree three [5]. We rederive the elementary algebraic facts needed here, rather than treating every neighboring research report as an axiom.

[write] *Notation against the rest of ProveIt.* The polynomial written H in (1.1) is the one the Algebra/JacobianConjecture project and the sibling reports [2, 3] write $h = u^2z + y^2(1 + 3u)$. In this article h is reserved for the *shear polynomial* $h(t)$ of (1.5), as in the source shears T_h of [2]; H never denotes a shear, and h never denotes the component factor. The inverse parameter $\tau = y + 1/x$ of theorem 2.2 is the variable that [3] writes t , a root of the same cubic (2.6); it is not the shear invariant xy , which the project's research notes and [2] write t . The letter L is used locally three times, each defined where it occurs: the field $\mathbb{K}(P, Q, R)$ in the proof of theorem 2.2, the linear form $xz + 3y$ of (4.5), and a vector of logarithms in section 7. The map (1.2) is the Lean definition counterexample (Algebra/JacobianConjecture/Lean/JacobianConjecture/Counterexample.lean), term for term. Its determinant is the Lean theorem jacobianDet_counterexample, over every commutative ring, and the collision (2.4) is collision, assembled from collision₀_value and collision₁_value; the Rocq counterparts in Coq/Counterexample.v are jacobian_det_is_minus_two and integral_collision_0_value, integral_collision_1_value. These are the only statements of this article that ProveIt has formalized.

Giannini's Zenodo version 1.1 record, dated 20 July 2026, lists the first six positive-iterate degrees

$$7, 43, 265, 1633, 10063, 62011,$$

and suggests $3 + \sqrt{10}$ as the dynamical degree [4]. A finite sequence alone does not prove that assertion: cancellation might occur later. The central question is whether one can control the leading terms for every iterate. We answer that question affirmatively and make the control uniform in positive weights and polynomial source shears.

1.2 Main conclusions

Write $F^n = F \circ \cdots \circ F$ (n factors), $F^0 = \text{Id}$, and $d_n = \deg(F^n)$. For a dominant polynomial map G , define

$$\lambda_1(G) = \lim_{n \rightarrow \infty} \deg(G^n)^{1/n}. \quad (1.3)$$

The usual existence follows by applying subadditivity to $\log \deg(G^n)$; our formulas prove existence directly in every case studied below.

Theorem 1.1 (Main degree statements). *Over any field of characteristic different from three, the map (1.2) has*

$$d_0 = 1, \quad d_1 = 7, \quad d_{n+2} = 6d_{n+1} + d_n, \quad \lambda_1(F) = 3 + \sqrt{10}. \quad (1.4)$$

Let $0 \neq h \in \mathbb{K}[t]$, $m = \deg h \geq 0$, and set

$$T_h(x, y, z) = (x, y, z + y^2 h(xy)), \quad F_h = F \circ T_h. \quad (1.5)$$

For $e_n = \deg(F_h^n)$ one has

$$\begin{aligned} e_0 &= 1, & e_1 &= 2m + 8, \\ e_{n+2} &= (2m + 7)e_{n+1} + (m + 3)e_n, & \lambda_1(F_h) &= \frac{2m + 7 + \sqrt{4m^2 + 32m + 61}}{2}. \end{aligned} \quad (1.6)$$

In characteristic different from two these maps have determinant -2 and separable generic degree three. In characteristic two the degree formulas still hold, but the Keller condition does not.

The proof is distributed across [sections 2, 3 and 5](#). No numerical extrapolation enters it. Two further conclusions make the calculation more than a single recurrence: [section 4](#) computes the upper Newton geometry of every iterate, and [section 7](#) turns the same matrices into an analytic escape theorem and an arithmetic-degree statement for a Zariski-dense set of initial points.

1.3 What is inherited and what is established here

The distinction matters because the repository contains both kernel-checked mathematics and unrefereed research reports. The classification report [\[2\]](#) already constructs the shears (1.5), describes a much larger normalized weighted class, and gives its ordinary-degree spectrum. The arithmetic report [\[3\]](#) already uses a cubic inverse parameter. We do not claim those constructions, the original counterexample, or its generic degree as new.

Object or assertion	Role in this article
F , determinant, collision	Known baseline; rechecked and credited.
T_h and weighted classification	Known family; no new classification claimed.
$3 + \sqrt{10}$	Previously suggested value; an all-iterate proof is given.
Weighted degrees and wall forms	Exact theorems proved here for every iterate.
Shear spectrum and characteristic three	Uniform formulas proved here.
Escape asymptotics and arithmetic points	Proved here under explicit spectral hypotheses.

“Proved here” describes the contents of this article, not an assertion that no prior author has obtained the result. The public record underlying the numerical prediction was accessible; its linked PDF was not successfully retrieved. Consequently, we neither assign a conjecture number to that manuscript nor assert that the prediction remained unproved in all literature. The source ledger in [section A](#) records this boundary explicitly.

2 Algebraic baseline and generic degree

2.1 Expansion and a localized Jacobian computation

The full expansion is

$$P = z + 3xyz + 3x^2y^2z + x^3y^3z + 4y^2 + 7xy^3 + 3x^2y^4, \quad (2.1)$$

$$Q = y + 3xz + 6x^2yz + 3x^3y^2z + 12xy^2 + 9x^2y^3, \quad (2.2)$$

$$R = 2x - 3x^2y - x^3z. \quad (2.3)$$

These formulas will provide a finite support certificate for all later degree arguments.

Proposition 2.1 (Known identities, rederived). *As an identity over $\mathbb{Z}[x, y, z]$, $\det J_F = -2$. Moreover,*

$$F(-1, 1, 5) = F(0, -2, -16) = (0, -2, 0). \quad (2.4)$$

Thus over a field of characteristic different from two, F is a noninjective Keller map.

Proof. The collision is direct substitution. For the determinant, work first in $\mathbb{Q}(x, y, z)$ and use the intermediate coordinates (x, u, H) from (1.1). Their Jacobian determinant is xu^2 . In these coordinates the target is

$$\left(uH, \frac{u-1}{x} + 3xH, \frac{x(u+1-x^2H)}{u^2} \right). \quad (2.5)$$

Indeed,

$$u^2(5-3u) + (u-1)^2(1+3u) = u+1.$$

Differentiating (2.5) with respect to (x, u, H) gives determinant $-2/(xu^2)$. Multiplication by xu^2 proves the claim in the rational function field, hence in the polynomial ring. Since both sides have integral coefficients, it is an identity over $\mathbb{Z}$ and specializes to every characteristic. The two source points in (2.4) remain distinct over every field, since their first coordinates differ by one. $\square$

2.2 The cubic parameter and what it does not measure

The next calculation is included to make the comparison between first and generic degree self-contained. The inverse parameter is the one used in [3]; it is not the invariant xy occurring in the shear.

Proposition 2.2 (Generic degree). *If $\text{char } \mathbb{K} \neq 2$, then F is dominant and*

$$[\mathbb{K}(x, y, z) : \mathbb{K}(P, Q, R)] = 3,$$

with separable extension. Every F_h has the same property, and its n th iterate has function-field degree 3^n .

Proof. Put $\tau = y + 1/x$, and for formal target coordinates (A, B, C) write

$$g_{A,B,C}(T) = CT^3 - 2T^2 + BT - 2A. \quad (2.6)$$

Expansion using (2.1)–(2.3) proves

$$g_{P,Q,R}(\tau) = 0, \quad g'_{P,Q,R}(\tau) = 2/x. \quad (2.7)$$

The second identity supplies the rational reconstruction

$$x = \frac{2}{g'(\tau)}, \quad y = \tau - \frac{1}{x}, \quad z = \frac{2 - 3xy - C/x}{x^2}. \quad (2.8)$$

Here and below identities involving $1/x$ are identities of rational function fields, not claims about every special fiber.

Let $L = \mathbb{K}(P, Q, R)$. Equations (2.7) and (2.8) show $\mathbb{K}(x, y, z) = L(\tau)$ and $[L(\tau) : L] \leq 3$. Thus $\text{trdeg}_{\mathbb{K}} L = 3$, so P, Q, R are algebraically independent and F is dominant. The polynomial (2.6) is irreducible in $\mathbb{K}[A, B, C, T]$: it is linear in A with invertible coefficient -2 , and its quotient ring is $\mathbb{K}[B, C, T]$. As a polynomial in T it is primitive over $\mathbb{K}[A, B, C]$, since its T^2 coefficient is the unit -2 . Gauss's lemma makes it irreducible over $\mathbb{K}(A, B, C)$. Its derivative does not vanish at τ by (2.7); hence the extension is separable of degree three.

The inverse of T_h is T_{-h} . Composition with this polynomial automorphism preserves generic degree and separability. Finally, the pullback by a dominant map identifies each successive pair in the function-field tower with the first pair, so the tower law gives 3^n . $\square$

Remark 2.3. Over characteristic zero the top dynamical degree is therefore three. It must not be substituted for $\lambda_1 = 3 + \sqrt{10}$. Nor does the number of points of a particular exceptional fiber determine either of these quantities. This article does not compute the second dynamical degree.

[write] Within ProveIt, the cubic (2.6) and the identities (2.7) are the inverse cubic and the derivative identity $g'(t) = 2/x$ of [3], where they are credited in turn to earlier external work; generic degree three of F is Gao's [5]. Theorem 2.2 is therefore a second route to a known statement for F ; what it adds is the passage to the shears F_h and to the iterates.

2.3 The exceptional algebra in characteristic two

Although characteristic two is not a Keller case, it is useful to separate its algebraic degeneration from the different dynamical degeneration at three.

Proposition 2.4. *In characteristic two the map F is dominant of purely inseparable function-field degree two. Its total-degree growth remains (1.4).*

Proof. The degree assertion is proved in section 3; its leading coefficients remain nonzero in characteristic two. For dominance and the field extension, write $A = P, B = Q, C = R$ and $\tau = y + 1/x$. Reduction of the polynomial identities gives

$$B = C\tau^2, \quad A = \tau(B + \tau + 1/x).$$

Consequently

$$\frac{1}{x} = \frac{A}{\tau} + B + \tau, \quad y = \tau + \frac{1}{x}, \quad z = \frac{C + x^2y}{x^3}.$$

It follows that $\mathbb{K}(x, y, z) = \mathbb{K}(A, C, \tau)$. These three generators are algebraically independent because the left side has transcendence degree three. The subfield $\mathbb{K}(A, C, C\tau^2)$ is rational of the same transcendence degree, and B/C is not a square in $\mathbb{K}(A, B, C)$, as its valuation at the prime B is one. Thus adjoining τ is a purely inseparable quadratic extension. The determinant is zero by theorem 2.1; no Keller conclusion is intended. $\square$

3 Every positive weighted degree and the exact recurrence

3.1 A composition lemma that rules out cancellation

For $w = (a, b, c) \in \mathbb{R}_{>0}^3$ and a nonzero polynomial $f = \sum_{\alpha} f_{\alpha} x^{\alpha_1} y^{\alpha_2} z^{\alpha_3}$, define

$$\deg_w f = \max_{f_{\alpha} \neq 0} \alpha \cdot w, \quad \text{in}_w f = \sum_{\alpha \cdot w = \deg_w f} f_{\alpha} X^{\alpha}.$$

These definitions allow arbitrary positive real weights, not just integers. Products of nonzero polynomials satisfy $\deg_w(fg) = \deg_w f + \deg_w g$: the product of their initial forms is nonzero in the polynomial ring over a field.

Lemma 3.1 (Unique outer leading monomial). *Let $G = (G_1, G_2, G_3)$ have nonzero polynomial components, with weighted degree vector $d = (\deg_w G_1, \deg_w G_2, \deg_w G_3)$. Suppose a polynomial H_i has exactly one monomial $c_i X^{\alpha_i}$ of maximal d -weight. Then*

$$\deg_w H_i(G) = \alpha_i \cdot d,$$

$$\text{in}_w H_i(G) = c_i \prod_{j=1}^3 (\text{in}_w G_j)^{\alpha_{ij}}.$$

In particular, there is no requirement that the initial forms of the G_j themselves be monomials.

Proof. The displayed product is nonzero and has the asserted weight. Every other outer monomial has strictly smaller d -weight, hence after substitution strictly smaller w -degree. It cannot cancel any term of the displayed product. $\square$

The uniqueness requirement concerns the *outer* polynomial. This distinction is the reason the argument below also works on the wall where each initial component of F has two terms.

3.2 Two matrices and a one-step invariant cone

Set

$$M = \begin{pmatrix} 3 & 3 & 1 \\ 3 & 2 & 1 \\ 3 & 0 & 1 \end{pmatrix}, \quad N = \begin{pmatrix} 2 & 4 & 0 \\ 2 & 3 & 0 \\ 2 & 1 & 0 \end{pmatrix}, \quad \mathbf{1} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}. \quad (3.1)$$

The rows of M give the exponents of $(x^3 y^3 z, 3x^3 y^2 z, -x^3 z)$; those of N give the exponents of $(3x^2 y^4, 9x^2 y^3, -3x^2 y)$. Assume throughout this section that $\text{char } \mathbb{K} \neq 3$, so these six coefficients are nonzero.

Lemma 3.2 (Support and cone certificate). *For $w = (a, b, c) > 0$ write $\delta(w) = a - b + c$. Then*

$$T(w) := \begin{pmatrix} \deg_w P \\ \deg_w Q \\ \deg_w R \end{pmatrix} = Nw + \max\{\delta(w), 0\} \mathbf{1}. \quad (3.2)$$

If $\delta(w) > 0$, the maximal monomial in each component is the one encoded by M ; if $\delta(w) < 0$, it is the one encoded by N . In all cases $\delta(T(w)) > 0$.

Proof. Inspection of (2.1)–(2.3) shows that every exponent in the i th component is coordinatewise bounded by either the i th row of M or the i th row of N . Apart from these two endpoints, each such bound is strict in at least one coordinate. Positive weights therefore put all other exponents strictly below the larger endpoint. The endpoint difference is

$$M - N = \mathbf{1} \begin{pmatrix} 1 & -1 & 1 \end{pmatrix},$$

which proves (3.2). Finally,

$$\delta(Mw) = 3a + b + c > 0,$$

$$\delta(Nw) = 2a + 2b > 0.$$

At $\delta(w) = 0$ the two vectors agree, and the conclusion is the same. $\square$

Theorem 3.3 (All positive weighted degrees). *For every $w \in \mathbb{R}_{>0}^3$ and $n \geq 1$,*

$$D_n(w) := \begin{pmatrix} \deg_w(F^n)_1 \\ \deg_w(F^n)_2 \\ \deg_w(F^n)_3 \end{pmatrix} = M^{n-1}(Nw + \max\{a - b + c, 0\}\mathbf{1}). \quad (3.3)$$

Proof. The first step is [theorem 3.2](#). Its degree vector belongs to the positive cone $\delta > 0$, where each component of the next outer copy of F has a unique maximal monomial encoded by M . Apply [theorem 3.1](#) to obtain $D_{n+1} = MD_n$. The same cone lemma keeps every subsequent vector in $\delta > 0$, completing the induction. $\square$

This proof uses only a finite support comparison, two strict linear inequalities, and an induction. In particular, it proves that no later cancellation can invalidate the observed degree pattern.

3.3 Ordinary degrees, generating functions, and a closed form

For ordinary degree $w = \mathbf{1}$, the initial gap is $1 > 0$, so

$$D_n(\mathbf{1}) = M^n \mathbf{1} \quad (n \geq 0). \quad (3.4)$$

The resulting vectors begin as follows.

n	$\deg(F^n)_1$	$\deg(F^n)_2$	$\deg(F^n)_3$
0	1	1	1
1	7	6	4
2	43	37	25
3	265	228	154
4	1633	1405	949
5	10063	8658	5848
6	62011	53353	36037

The first coordinate is the largest for $n \geq 1$: the differences between the first and second rows of M , and between the second and third rows, are $(0, 1, 0)$ and $(0, 2, 0)$ respectively.

Theorem 3.4 (Exact scalar laws). *Let $d_n = \deg F^n$. Then*

$$\sum_{n \geq 0} d_n z^n = \frac{1+z}{1-6z-z^2}. \quad (3.5)$$

Writing $s = \sqrt{10}$ and $\lambda_{\pm} = 3 \pm s$, one has

$$d_n = \frac{(s+4)\lambda_+^n + (s-4)\lambda_-^n}{2s}. \quad (3.6)$$

In particular $\lambda_1(F) = \lambda_+$.

Proof. Direct calculation gives

$$\det(tI - M) = t(t^2 - 6t - 1), \quad \text{im } M = E := \{(a, b, c) : 2a - 3b + c = 0\}.$$

The matrix has rank two. On E it satisfies $M^2 = 6M + I$; this follows either by restriction or from $M(M^2 - 6M - I) = 0$ and the fact that M is invertible on its image. Since $\mathbf{1} \in E$, every coordinate of $M^n \mathbf{1}$ satisfies the recurrence from its initial index $n = 0$. For the first coordinate the initial values are 1, 7, giving (3.5) and then (3.6). The coefficient of λ_+^n is positive, $\lambda_+ > 1$, and $|\lambda_-| < 1$, so taking n th roots proves the limit. $\square$

For completeness, the other two coordinate generating functions are

$$\sum_{n \geq 0} \deg(F^n)_2 z^n = \frac{1}{1 - 6z - z^2}, \quad \sum_{n \geq 0} \deg(F^n)_3 z^n = \frac{1 - 2z}{1 - 6z - z^2}.$$

The small negative conjugate $3 - \sqrt{10}$ will reappear in the transverse contraction governing escape to infinity.

4 Upper Newton geometry and exact wall forms

4.1 What is computed

The Newton polytope of a polynomial is the convex hull of all its exponent vectors. Positive weighted degrees see only its upper faces. It is useful to encode exactly that information by the *downward completion*

$$\mathcal{P}^\downarrow(f) = \text{Newt}(f) - \mathbb{R}_{\geq 0}^3. \quad (4.1)$$

This is not the full Newton polytope. Its support function is finite on the nonnegative orthant and agrees there with that of $\text{Newt}(f)$. Calling it an upper Newton object refers to its maximal faces, even though its completion extends downward.

Theorem 4.1 (Two vertices determine the upper geometry). *Assume $\text{char } \mathbb{K} \neq 3$. For $n \geq 1$ and $i \in \{1, 2, 3\}$ put*

$$v_{n,i}^+ = (M^n)_{i,*}, \quad v_{n,i}^- = (M^{n-1}N)_{i,*}, \quad k_n = M^{n-1}\mathbf{1}. \quad (4.2)$$

Then

$$\mathcal{P}^\downarrow((F^n)_i) = \text{conv}\{v_{n,i}^+, v_{n,i}^-\} - \mathbb{R}_{\geq 0}^3, \quad (4.3)$$

and

$$v_{n,i}^+ - v_{n,i}^- = k_{n,i}(1, -1, 1). \quad (4.4)$$

Thus the only wall for strictly positive weights is $a + c = b$, independently of the iterate.

Proof. By [theorem 3.3](#), the support function for positive weights is

$$(M^{n-1}Nw)_i + k_{n,i} \max\{a - b + c, 0\} = \max\{v_{n,i}^+ \cdot w, v_{n,i}^- \cdot w\}.$$

Here $k_{n,i} > 0$. Off the wall, successive applications of [theorem 3.1](#) show that the initial form is a nonzero monomial of exponent $v_{n,i}^+$ or $v_{n,i}^-$ respectively. Both endpoints therefore occur in the polynomial. Equality of the support functions extends to nonnegative weights by continuity. A closed downward-completed convex polyhedron is determined by these support values: a separating linear functional with finite upper bound must have nonnegative coefficients. This proves (4.3). Equation (4.4) follows by multiplying $M - N = \mathbf{1}(1, -1, 1)$ on the left by M^{n-1} . $\square$

An equivalent elementary version of the last separation step is the following. If a support exponent lay outside the right side of (4.3), a nonnegative weight would strictly separate it. Perturbing this weight slightly into the positive orthant would preserve strict separation, contradicting [theorem 3.3](#).

4.2 The entire initial form on the wall

Put

$$L = xz + 3y, \quad B_0 = \begin{pmatrix} 2 & 3 & 0 \\ 2 & 2 & 0 \\ 2 & 0 & 0 \end{pmatrix}. \quad (4.5)$$

The notation B_0 is an exponent matrix, not the second target coordinate from (2.6). For a positive wall weight $a + c = b$, the first initial forms are

$$\text{in}_w F = (x^2 y^3 L, 3x^2 y^2 L, -x^2 L). \quad (4.6)$$

They need not remain monomials, but their propagation is nevertheless exact.

Theorem 4.2 (Wall factorization at every iterate). *Let $C_1 = (1, 3, -1)$ and recursively define nonzero coefficients*

$$C_{n+1,i} = C_{1,i} \prod_{j=1}^3 C_{n,j}^{M_{ij}}. \quad (4.7)$$

If $w > 0$ and $a + c = b$, then for every $n \geq 1$,

$$\text{in}_w(F^n)_i = C_{n,i} x^{(M^{n-1}B_0)_{i1}} y^{(M^{n-1}B_0)_{i2}} L^{(M^{n-1}\mathbf{1})_i}. \quad (4.8)$$

Proof. The case $n = 1$ is (4.6). After that first step the degree vector lies strictly in the plus cone. Therefore the unique outer initial monomials have exponents $M_{i,*}$ and coefficients $C_{1,i}$. The composition lemma multiplies the exponent matrix by M , the exponent vector of L by M , and the coefficient vector according to (4.7). This is exactly the stated induction. The coefficients remain nonzero because $3 \neq 0$ and the coefficient ring is a field. $\square$

For example, $C_2 = (-27, -27, 1)$. Formula (4.8) describes all cancellation-sensitive leading terms along this wall. In positive characteristic some interior binomial coefficients in L^k can vanish, but neither endpoint of that power vanishes when $3 \neq 0$. The upper-geometry theorem is therefore unchanged.

4.3 A robust support class

The exact growth is not specific to all sixteen coefficients in (2.1)–(2.3). It survives arbitrary changes below the relevant upper faces.

Corollary 4.3 (Coefficient robustness). *Let $G = (G_1, G_2, G_3)$ be a polynomial tuple over any field. Suppose*

$$\text{supp}(G_i) \subseteq (\text{conv}\{M_{i,*}, N_{i,*}\} - \mathbb{R}_{\geq 0}^3) \cap \mathbb{N}^3$$

and the coefficient of $X^{M_{i,}}$ is nonzero for every i . Then the ordinary coordinate-degree vector of G^n is $M^n \mathbf{1}$. If G is dominant, $\lambda_1(G) = 3 + \sqrt{10}$.*

Proof. For every positive weight in the plus cone, $M_{i,*}$ is the unique maximizer on the specified support region. The ordinary initial weight belongs to that cone, and M preserves it. The proof of theorem 3.3 therefore applies without change. Dominance is needed only to use the conventional dynamical-degree terminology, not for the degree-vector induction. $\square$

In particular, in characteristic different from three the same law holds after independent invertible affine changes in each source coordinate and in each target coordinate, with no coordinate mixing or permutation. Expanding such a substitution moves exponents only downward from existing ones, and the plus vertex retains a nonzero coefficient. These source–target modifications need not be conjugacies. The result is a support statement, not an assertion that first dynamical degree is invariant under arbitrary left–right equivalence.

Lemma 4.4 (Weighted eigenvector principle). *More generally, suppose a polynomial tuple G has unique v -leading monomials $c_i X^{A_i,*}$, where $v > 0$, $A \in M_d(\mathbb{N})$, and $Av = \lambda v$ with $\lambda > 1$. Then*

$$\deg_v(G^n)_i = \lambda^n v_i.$$

Its ordinary degree has exponential rate λ .

Proof. The degree vector after each step is a positive scalar multiple of v , so the unique outer leading monomials remain unique. Induction using [theorem 3.1](#) proves the weighted formula. If $v_{\min} = \min_i v_i$ and $v_{\max} = \max_i v_i$, then

$$v_{\min} \deg f \leq \deg_v f \leq v_{\max} \deg f$$

for every nonzero polynomial f . These fixed multiplicative comparison constants disappear on taking n th roots. $\square$

This principle is a general leading-term method. Our contribution is the explicit invariant cones, matrices, and consequences for the present maps, rather than a claim to have invented weighted degrees.

5 The complete polynomial-shear spectrum

5.1 Why a shear is dynamically nontrivial

The transformation T_h in (1.5) is triangular, with Jacobian one and inverse T_{-h} . Thus $F_h = F \circ T_h$ is left-right equivalent to F . It is usually not conjugate to F : iteration produces

$$F_h^n = (F \circ T_h) \circ \cdots \circ (F \circ T_h),$$

not $F^n \circ T_h$ and not $T_h^{-1} \circ F^n \circ T_h$. This is the distinction that permits unbounded dynamical degrees. The repository already uses these shears in its weighted classification [2]; here they are treated as dynamical systems in the fixed displayed coordinates.

Let $0 \neq h(t) \in \mathbb{K}[t]$, let $m = \deg h$, and write $\eta = \text{lc}(h)$. Substitution in (1.2) gives

$$\begin{aligned} P_h &= P + (1 + xy)^3 y^2 h(xy), \\ Q_h &= Q + 3x(1 + xy)^2 y^2 h(xy), \\ R_h &= R - x^3 y^2 h(xy). \end{aligned} \tag{5.1}$$

In characteristic different from three the candidate top monomials are

$$\eta x^{m+3} y^{m+5}, \quad 3\eta x^{m+3} y^{m+4}, \quad -\eta x^{m+3} y^{m+2}. \tag{5.2}$$

Their exponent matrix is

$$S_m = \begin{pmatrix} m+3 & m+5 & 0 \\ m+3 & m+4 & 0 \\ m+3 & m+2 & 0 \end{pmatrix}. \tag{5.3}$$

Lemma 5.1 (Shear cone). *For positive weights $w = (a, b, c)$ put*

$$\Delta_m(w) = ma + (m+2)b - c.$$

If $\Delta_m(w) > 0$, the monomials in (5.2) are the unique maximal monomials of the corresponding components. Moreover,

$$\Delta_m(S_m w) = (2m+1)(m+3)a + (2m^2 + 10m + 6)b > 0. \tag{5.4}$$

Proof. Every exponent in the added shear part of a component is coordinatewise bounded by the corresponding row of S_m . Only the highest-degree term of h and the highest power of xy give that row. The difference between a row of S_m and the corresponding row of M is $(m, m+2, -1)$; the difference from the row of N is $(m+1, m+1, 0)$. The first difference has positive w -weight by hypothesis, and the second does so because $w > 0$. The support certificate for F now proves uniqueness. Direct multiplication gives (5.4). $\square$

Theorem 5.2 (All-iterate shear degree law). *Assume $\text{char } \mathbb{K} \neq 3$ and $h \neq 0$ has degree $m \geq 0$. For every $n \geq 0$,*

$$\begin{pmatrix} \deg(F_h^n)_1 \\ \deg(F_h^n)_2 \\ \deg(F_h^n)_3 \end{pmatrix} = S_m^n \mathbf{1}. \quad (5.5)$$

The ordinary degrees satisfy (1.6), and

$$\sum_{n \geq 0} \deg(F_h^n) z^n = \frac{1+z}{1-(2m+7)z-(m+3)z^2}. \quad (5.6)$$

They depend on h only through its degree, not through its other coefficients or its nonzero leading coefficient.

Proof. The ordinary initial weight satisfies $\Delta_m(\mathbf{1}) = 2m+1 > 0$. Apply theorems 3.1 and 5.1 inductively to obtain (5.5). The first coordinate is the largest at every positive step, since consecutive row differences are $(0, 1, 0)$ and $(0, 2, 0)$.

One has

$$\det(tI - S_m) = t(t^2 - (2m+7)t - (m+3)).$$

The image is the same plane $E = \{2a - 3b + c = 0\}$ as before. Its rank is two, and on E the matrix satisfies $S_m^2 = (2m+7)S_m + (m+3)I$. Since $\mathbf{1} \in E$, this gives the second-order recurrence from $n = 0$. The initial vector is $(2m+8, 2m+7, 2m+5)$, yielding (5.6).

The two nonzero eigenvalues are

$$\lambda_m = \frac{2m+7 + \sqrt{4m^2 + 32m + 61}}{2}, \quad \mu_m = -\frac{m+3}{\lambda_m}.$$

They satisfy $\lambda_m > m+3$ and $-1 < \mu_m < 0$. The scalar recurrence has a positive coefficient of λ_m^n , because $e_0 = 1$, $e_1 = 2m+8 > 0$, and $\mu_m < 0$. Its exponential rate is therefore λ_m . $\square$

Shear	$\deg F_h$	$\deg F_h^2$	$\deg F_h^3$	$\lambda_1(F_h)$
$h = 0$	7	43	265	$3 + \sqrt{10}$
$\deg h = 0$	8	59	437	$(7 + \sqrt{61})/2$
$\deg h = 1$	10	94	886	$(9 + \sqrt{97})/2$
$\deg h = 2$	12	137	1567	$(11 + \sqrt{141})/2$
$\deg h = 3$	14	188	2528	$(13 + \sqrt{193})/2$

The zero polynomial is a separate case, not a polynomial of degree zero. In particular, taking a nonzero constant shear coefficient arbitrarily close to zero over $\mathbb{C}$ does not preserve the first dynamical degree at the limiting zero coefficient.

[write] The case $n = 1$ of theorem 5.2, the coordinate degrees $(2m+8, 2m+7, 2m+5)$ of F_h (and $(7, 6, 4)$ for $h = 0$), is not new: it is Corollary 6.4 (“An exact degree spectrum”) of [2], stated there for the normalized maps $F_{a,b} \circ T_h$. These differ from $F \circ T_{h'}$ by diagonal scalings of source and target that replace h by a nonzero multiple of $h(ct)$, $c \neq 0$, so the degrees agree. What is new here is the statement for every iterate $n \geq 2$ and the resulting value of $\lambda_1(F_h)$.

5.2 Fixed generic degree, unbounded dynamical degree

Corollary 5.3. *Over characteristic zero, one tame left–right equivalence class contains noninjective Keller maps of separable generic degree three whose first dynamical degrees are unbounded. More precisely, the subfamily $\{F_h : h \in \mathbb{K}[t]\}$ realizes exactly*

$$\{3 + \sqrt{10}\} \cup \left\{ \frac{2m+7 + \sqrt{4m^2 + 32m + 61}}{2} : m \in \mathbb{N} \right\}. \quad (5.7)$$

Proof. Triangularity of T_h gives $\det J_{F_h} = -2$; generic degree is three by [theorem 2.2](#). The collision pulls back to

$$\begin{aligned} F_h(-1, 1, 5 - h(-1)) &= F_h(0, -2, -16 - 4h(0)) \\ &= (0, -2, 0). \end{aligned}$$

These source points are distinct. The spectrum follows from [theorems 3.4](#) and [5.2](#); its second family is unbounded because $\lambda_m > 2m + 7$. $\square$

This does not classify dynamical degrees in the full weighted class, still less in an entire arbitrary left–right orbit. Diagonal coordinate scalings preserve the relevant supports, but permutations and coordinate mixing can change the iteration mechanism. The precise statement of [theorem 5.3](#) is an existence result inside one equivalence class and an exact spectrum for the specified source-shear subfamily.

[write] The Keller property and the collision of F_h in the proof of [theorem 5.3](#) are those of [\[2\]](#), whose Theorem 6.3 transports collision points by T_{-h} ; the new content of the corollary is the unbounded spectrum ([5.7](#)).

6 A complete degree-growth phase change in characteristic three

All assertions in this section concern iteration in the polynomial ring over a field of characteristic three. They do not concern the finite set map obtained by evaluating on a finite field and reducing polynomial functions modulo $X^q - X$.

6.1 A simplifying coordinate identity

Reduction of the components gives

$$Q = y, \quad R = -x - x^3z, \quad P + y^3R = z + y^2. \quad (6.1)$$

For the shear family the identities become

$$\begin{aligned} Q_h &= y, \\ R_h &= -x - x^3z - x^3y^2h(xy), \\ P_h + y^3R_h &= z + y^2(1 + h(xy)). \end{aligned} \quad (6.2)$$

Thus the second coordinate is fixed under iteration. These exact identities replace the matrix M , whose second leading coefficient has disappeared.

Theorem 6.1 (Characteristic-three phase diagram). *Let $\text{char } \mathbb{K} = 3$. For $n \geq 1$ the coordinate-degree vectors are as follows.*

(i) For $h = 0$,

$$(\deg(F^n)_1, \deg(F^n)_2, \deg(F^n)_3) = (7 \cdot 4^{n-1}, 1, 7 \cdot 4^{n-1} - 3). \quad (6.3)$$

(ii) For nonzero constant h ,

$$(\deg(F_h^n)_1, \deg(F_h^n)_2, \deg(F_h^n)_3) = (8 \cdot 4^{n-1}, 1, 8 \cdot 4^{n-1} - 3). \quad (6.4)$$

(iii) For $\deg h = m \geq 1$,

$$(\deg(F_h^n)_1, \deg(F_h^n)_2, \deg(F_h^n)_3) = (a_n, 1, a_n - 3), \quad (6.5)$$

where

$$a_1 = 2m + 8, \quad a_{n+1} = (m + 3)a_n + m + 5, \quad (6.6)$$

$$a_n = \left(2m + 8 + \frac{m + 5}{m + 2}\right) (m + 3)^{n-1} - \frac{m + 5}{m + 2}. \quad (6.7)$$

Consequently

$$\lambda_1(F_h) = \begin{cases} 4, & h = 0 \text{ or } \deg h = 0, \\ m + 3, & \deg h = m \geq 1. \end{cases} \quad (6.8)$$

Proof. For $h = 0$, the first degree vector is $(7, 1, 4)$. For $h \neq 0$ of degree $m \geq 0$, it is $(2m + 8, 1, 2m + 5)$: the top monomials of P_h and R_h have coefficients η and $-\eta$, which do not vanish. Suppose at some positive step the degrees are $(a, 1, a - 3)$.

In the next third coordinate, the term $-x^3z$ has substituted degree $4a - 3$. For a nonzero shear the top term $-\eta x^{m+3}y^{m+2}$ has substituted degree $(m + 3)a + m + 2$. If $m = 0$, the difference in favor of $-x^3z$ is $a - 5 > 0$, since $a \geq 8$. If $m \geq 1$, the difference in favor of the shear is

$$(m - 1)a + m + 5 > 0.$$

Each winning term is unique. For $h = 0$ only the first comparison is needed, and the linear term $-x$ is smaller.

Use the last identity in (6.2) after substitution. The term $-y^3R_h$ has degree three more than the new third coordinate. The right-side terms have degrees at most $a - 3$ and, when $h \neq 0$, $ma + m + 2$. They are strictly smaller than that new first degree. Therefore the new first degree is exactly three more than the new third degree. This proves preservation of the shape $(a, 1, a - 3)$.

The winning term gives $a' = 4a$ for $h = 0$ and for nonzero constant h , and $a' = (m + 3)a + m + 5$ for $m \geq 1$. The initial values prove (6.3)–(6.6). Solving the last linear recurrence over the rational numbers yields (6.7), and taking n th roots gives (6.8). $\square$

The rational fractions in (6.7) are expressions for ordinary integer degree sequences. They are not field elements in characteristic three; no inversion of $m + 2$ in $\mathbb{K}$ is used. For example, when $m = 1$ the formula is $a_n = 12 \cdot 4^{n-1} - 2$.

Corollary 6.2 (Reduction of an integral shear). *For a fixed $h \in \mathbb{Z}[t]$, reducing modulo three and computing the actual polynomial $\bar{h} \in \mathbb{F}_3[t]$ determines the degree law completely by theorem 6.1. Its degree can be smaller than $\deg h$, and $\bar{h}$ can be zero. At any prime different from three where the leading coefficient survives, theorem 5.2 applies with the original degree.*

The determinant remains nonzero in characteristic three, and theorem 2.2 still gives separable generic degree three. Thus the change from $3 + \sqrt{10}$ to 4 is a genuine change in iteration complexity, not the loss of dominance or of the Keller condition.

7 Escape at infinity and arithmetic degree

The degree matrices also control actual orbits whose coordinate magnitudes lie in a suitable logarithmic cone. A leading monomial is then not merely largest in a grading: it dominates all other terms in absolute value, uniformly in their complex arguments. The small nonleading eigenvalues of the matrices make the transverse errors contract.

7.1 A general monomial escape theorem

For $z = (z_1, \dots, z_d) \in (\mathbb{C}^*)^d$ set

$$\mathcal{L}(z) = (\log |z_1|, \dots, \log |z_d|)^\top.$$

The following theorem is stated in a dimension-independent form so that its hypotheses and limitations are visible.

Theorem 7.1 (Monomial dominance with contracting transverse spectrum). *Let $G = (G_1, \dots, G_d)$ be a complex polynomial tuple. Suppose there exist a matrix $A \in M_d(\mathbb{N})$, a vector $v \in \mathbb{R}_{>0}^d$, a number $\lambda > 1$, and coefficients $c_i \neq 0$ such that:*

- (a) $Av = \lambda v$, and the unique v -leading monomial of G_i is $c_i X^{A_{i,*}}$;
- (b) λ is a simple eigenvalue of A , and every other eigenvalue has modulus less than one;
- (c) there is a nonnegative left eigenfunctional ℓ with $\ell A = \lambda \ell$ and $\ell(v) = 1$.

Put $b = (\log |c_1|, \dots, \log |c_d|)^\top$ and

$$L_* = (I - A)^{-1}b. \quad (7.1)$$

Then there is a nonempty forward-invariant open set $\mathcal{U} \subset (\mathbb{C}^*)^d$ containing all points with logarithms sufficiently far out in a sufficiently narrow cone around $L_* + \mathbb{R}_{>0}v$. There is a continuous positive function $\mathcal{G} : \mathcal{U} \rightarrow \mathbb{R}_{>0}$ such that

$$\mathcal{G}(Gz) = \lambda \mathcal{G}(z). \quad (7.2)$$

If ρ is the maximum modulus of the other eigenvalues (zero when there are none), then for any q with $\rho < q < 1$ and every $z \in \mathcal{U}$,

$$\mathcal{L}(G^n z) = \lambda^n \mathcal{G}(z)v + L_* + O_z(q^n). \quad (7.3)$$

The error can be taken uniform for z in a fixed compact subset of $\mathcal{U}$.

Proof. Since no eigenvalue equals one, L_* is defined. Let $V = \ker \ell$. The restriction $A|_V$ has spectral radius $\rho < 1$. Choose $r_0 \in (\rho, 1)$ and a norm on V with operator norm at most r_0 . This choice is fixed for the construction of $\mathcal{U}$. Such a norm exists: for example, start with any norm and use a convergent weighted sum of $\|(A|_V)^j e\|$ with ratio chosen strictly between ρ and r_0 .

Write logarithmic vectors uniquely as

$$L = L_* + tv + e, \quad t = \ell(L - L_*), \quad e \in V. \quad (7.4)$$

For each nonleading exponent α of G_i there is a strict gap

$$(A_{i,*} - \alpha) \cdot v > 0.$$

There are finitely many such gaps. Hence for sufficiently small $\varepsilon > 0$, whenever $\|e\| < \varepsilon t$ all the expressions

$$(A_{i,*} - \alpha) \cdot (tv + e)$$

are at least $\kappa_0 t$ for a fixed $\kappa_0 > 0$. Constants from L_* and the coefficients can be absorbed into a common bound. Uniformly in the arguments of the z_j , the ratio of each nonleading term to the leading term is therefore $O(e^{-\kappa_0 t})$.

Choose R so large that these ratios have sum less than $1/2$ in each component when $t > R$. Then no component of Gz vanishes, and

$$\mathcal{L}(Gz) = A\mathcal{L}(z) + b + E(z), \quad \|E(z)\| \leq Ce^{-\kappa t} \quad (7.5)$$

for fixed positive C, κ . This follows from $\log |1 + r| = O(|r|)$ for $|r| < 1/2$; it does not require choosing a complex logarithm.

Since $AL_* + b = L_*$, the coordinates of the image in (7.4) satisfy

$$t' = \lambda t + \ell E(z), \quad e' = Ae + (I - v\ell)E(z). \quad (7.6)$$

Fix $1 < \sigma < \lambda$. By increasing R , the exponential error makes $t' \geq \sigma t$ and $\|e'\| < \varepsilon t'$ whenever $t > R$ and $\|e\| < \varepsilon t$. Indeed, the linear contribution to $\|e'\|$ is at most $r_0 \varepsilon t$, whereas $\varepsilon t'$ is asymptotic to $\varepsilon \lambda t$; the difference is a positive linear margin. Thus the set

$$\mathcal{U} = \{z \in (\mathbb{C}^*)^d : t(z) > R, \|e(z)\| < \varepsilon t(z)\} \quad (7.7)$$

is forward invariant. Shrink ε as needed to ensure that all logarithmic coordinates tend to $+\infty$ with t ; this is possible because $v > 0$.

For $z_n = G^n z$, let $t_n = t(z_n)$ and $E_n = E(z_n)$. Then $t_n \geq \sigma^n t_0$ and $\|E_n\| \leq C \exp(-\kappa \sigma^n t_0)$. Define

$$\mathcal{G}(z) = t_0 + \sum_{j=0}^{\infty} \lambda^{-j-1} \ell E_j. \quad (7.8)$$

The series converges uniformly on compact subsets of $\mathcal{U}$. It differs from t_0 by an exponentially small quantity as $t_0 \rightarrow \infty$, so enlarging R makes it positive. Iterating (7.6) shows that $\lambda^{-n} t_n \rightarrow \mathcal{G}(z)$ and proves (7.2). The tail estimate gives

$$t_n - \lambda^n \mathcal{G}(z) = O_z(\exp(-\kappa' \sigma^n))$$

for some $\kappa' > 0$.

Finally,

$$e_n = A^n e_0 + \sum_{j=0}^{n-1} (A|_V)^{n-1-j} (I - v\ell) E_j.$$

For any specified $q \in (\rho, 1)$ choose $r \in (\rho, q)$. The finite-dimensional spectral-radius estimate gives $\|(A|_V)^n\| \leq C_r r^n$ in the fixed norm. Since the forcing terms decay faster than every geometric sequence, the displayed sum is $O_z(r^n)$, hence $O_z(q^n)$. This argument does not change $\mathcal{U}$. Combining the last two estimates proves (7.3), with compact uniformity following from the same bounds. $\square$

7.2 Application to the baseline and the shears

Corollary 7.2 (Complex escape for the Keller family). *Over $\mathbb{C}$, both F and every F_h with $h \neq 0$ satisfy theorem 7.1, with exponent matrix M or S_m respectively and with the dynamical degrees computed above.*

Proof. For M take

$$v = \begin{pmatrix} (2 + \sqrt{10})/3 \\ (7 + 2\sqrt{10})/9 \\ 1 \end{pmatrix}. \quad (7.9)$$

It is positive and satisfies $Mv = (3 + \sqrt{10})v$. The matrix M^2 is strictly positive, so its Perron left eigenfunctional is positive; alternatively, it can be obtained by solving the three linear equations directly. The other eigenvalues are $3 - \sqrt{10}$ and zero, both of modulus less than one. The vector v lies in the plus cone because $\lambda \delta(v) = \delta(Mv) = 3v_1 + v_2 + v_3 > 0$. Thus the required leading monomials are exactly those encoded by M .

For S_m , the top-left 2×2 block is strictly positive. Its Perron eigenvector has positive entries, and the last row extends it to a positive right eigenvector of S_m . Its left Perron eigenfunctional has positive

first two entries and zero third entry. The remaining eigenvalues are $\mu_m \in (-1, 0)$ and zero. Equation (5.4), evaluated on a positive eigenvector, shows that $\Delta_m(v) > 0$. The unique leading monomials are consequently (5.2). All hypotheses of [theorem 7.1](#) hold. $\square$

For the baseline coefficients $(1, 3, -1)$, one obtains the particularly simple constant correction

$$L_* = (0, \frac{1}{2} \log 3, -\frac{3}{2} \log 3)^\top. \quad (7.10)$$

Thus, on the escape region, even the subexponential coordinate ratios are controlled after subtracting the leading ray. The asymptotic is stronger than merely saying that the orbit is unbounded.

For F_h , set $b_\eta = \log |\eta|$ and $s_3 = \log 3$. Solving $(I - S_m)L_* = (b_\eta, b_\eta + s_3, b_\eta)^\top$ gives

$$L_* = \frac{1}{3(m+3)} \begin{pmatrix} -2b_\eta - (m+5)s_3 \\ (m+2)s_3 - b_\eta \\ b_\eta - (4m+11)s_3 \end{pmatrix}. \quad (7.11)$$

Lower coefficients of h influence the escape function and the size of a valid region, but not the leading exponential rate or this constant logarithmic correction.

7.3 Arithmetic degree for many integral points

For an integral point $p \in \mathbb{Z}^d$, use the logarithmic projective height

$$h([1 : p_1 : \dots : p_d]) = \log \max\{1, |p_1|, \dots, |p_d|\}.$$

For an integral polynomial map G and an integral initial point, define the arithmetic degree, when the limit exists, by

$$\alpha_G(p) = \lim_{n \rightarrow \infty} \max\{1, h([1 : G^n p])\}^{1/n}. \quad (7.12)$$

It is the *logarithmic* height that appears under the n th root. Taking an n th root of the multiplicative height would be a different and generally divergent expression here.

Theorem 7.3 (A Zariski-dense set of maximal arithmetic degree). *Let $G = F$, or $G = F_h$ with $0 \neq h \in \mathbb{Z}[t]$. There is a Zariski-dense set of integral initial points p for which*

$$\alpha_G(p) = \lambda_1(G).$$

More precisely, every $p \in \mathcal{U} \cap \mathbb{Z}^3$ in the escape region from [theorem 7.2](#) has this property. With the same normalization of v and $\mathcal{G}$ as in [theorem 7.1](#),

$$h([1 : G^n p]) = v_1 \mathcal{G}(p) \lambda_1(G)^n + (L_*)_1 + O_p(q^n). \quad (7.13)$$

Proof. All iterates remain integral. For the positive right eigenvectors of both matrices, the first coordinate is strictly largest: subtracting the first and second eigenvector equations gives $\lambda(v_1 - v_2) = v_2 > 0$, and subtracting the second and third gives $\lambda(v_2 - v_3) = 2v_2 > 0$. By (7.3), the first logarithmic coordinate therefore eventually determines the maximum in the height. This proves (7.13). Its leading coefficient is positive, so (7.12) has limit $\lambda_1(G)$.

It remains to prove that the integral points under consideration are Zariski dense, rather than merely infinite. For sufficiently small $\eta_0 > 0$, every sufficiently large t has the logarithmic box

$$L_* + tv + [-\eta_0 t, \eta_0 t]^3$$

contained in the cone defining $\mathcal{U}$. This follows from the linear decomposition (7.4): the change in t and in the transverse part is bounded by a constant times $\eta_0 t$. Choose η_0 small enough to respect the cone and to keep $v_i - \eta_0 > 0$.

Exponentiating this box in the positive real octant gives a Cartesian box containing arbitrarily many distinct positive integers in each coordinate interval. A nonzero polynomial of degree at most D in each variable cannot vanish on a Cartesian grid with more than D distinct entries in each variable. This last assertion follows by induction on the number of variables, using the univariate root bound. Hence no nonzero polynomial can vanish on all of $\mathcal{U} \cap \mathbb{Z}^3$, proving Zariski density. $\square$

Remark 7.4 (The precise arithmetic boundary). A Zariski-dense *set of initial points* is not the same thing as a single Zariski-dense orbit. The theorem makes no assertion that every orbit starting in $\mathcal{U}$ is Zariski dense. It also does not assert the equality of arithmetic and dynamical degree for every rational point with a Zariski-dense orbit. It supplies an explicitly characterized escape region and proves the equality for all integral points in that region.

The construction is nonvacuous at the level of individual points: rounding each coordinate of $\exp(L_* + tv)$ to the nearest positive integer places the resulting point in $\mathcal{U}$ for all sufficiently large t , because the relative rounding errors tend to zero. The proof by boxes, rather than this single rounded ray, is what establishes Zariski density.

8 Verification, scope, and a formalization route

8.1 What the companion computation checks

The package contains `code/verify_results.py`, which was executed with Python 3.13.5 and SymPy 1.14.0. All thirteen check groups passed; the recorded output is in `data/verification.json` and `data/verification.txt`. The checks are exact, not floating-point fits. Their mathematical role is deliberately limited.

Group	Object checked	Scope
1	Jacobian and integral collision	Exact polynomial identities
2	Cubic equation and derivative identity	Rational-function identities
3	Characteristic-two inverse identities	Exact reduction modulo two
4	Support bounded by the two endpoints	Complete baseline support
5	First wall initial forms	Complete baseline support
6	Matrix polynomial and cone identities	Exact finite identities
7	Ordinary recurrence and degree vectors	First twelve steps
8	Weighted compositions over $\mathbb{F}_{101}$	Five weights, three steps
9	Shear characteristic polynomial and gap	Symbolic parameter m
10	Shear compositions over $\mathbb{F}_{101}$	Four coefficient lists
11	Characteristic-three degree laws	Six cases, four steps
12	Other small characteristics	Primes 2, 5, 7, 11
13	Spectral and fixed-point identities	Baseline; sample shear checks

The composition tests use exact univariate polynomial restrictions along monomial curves, not full expansion of every multivariate iterate. Such restrictions supply lower-bound certificates when they achieve a claimed degree; they can also lose leading terms at special curves. The selected tests avoid that loss. The upper bounds and the all-iterate statements come from the proofs, not from interpolation or from the finite list of tests.

Likewise, the escape theorem is proved analytically. No numerical orbit simulation is offered as its proof, and no empirical error bound is substituted for the finite-gap and contraction arguments. The general shear inequalities are proved for all m in the text; checking particular eigenvalues in the script does not replace that argument.

8.2 A short route to machine-checked degree growth

The core degree theorem has a relatively small logical footprint. It need not depend on the repository's classification results or on the theory of complex dynamics. A useful formalization order is as follows.

First formalize a support-based weighted degree over multivariate polynomials over a field. For the ordinary-degree theorem it is enough to use natural-number weights, finite supports, and the fact that nonzero products have additive degree. Next prove the unique-outer-leading-monomial composition lemma. This is the only place where a cancellation concern needs to be resolved at the polynomial level.

Then represent M, N, S_m as finite matrices and verify the endpoint support inequalities from the explicit component polynomials. The invariance of the plus cone and shear cone is elementary ordered arithmetic. Induction gives the degree vectors. The second-order recurrences can be derived directly from $(M^2 - 6M - I)\mathbf{1} = 0$ and the corresponding identity for S_m , followed by multiplication by matrix powers. This avoids needing a general spectral theorem in the first formalization pass.

A second pass can establish the generating functions, closed forms, and limits. A third pass can treat arbitrary real weights and convex separation for the upper Newton geometry. The escape theorem is naturally a separate analytic development involving norms, spectral contraction, and uniformly convergent series.

The inspected baseline Lean source includes `counterexample` and `collision`. Its determinant theorem is `jacobianDet_counterexample`. None of these supplies the new all-iterate statements in this package. No Lean or Rocq executable was run for this article, and no uncompiled source is presented as a formal proof.

[write] At the placement commit 51c6943bf these declarations are unchanged from the pin: `counterexample`, `jacobianDet_counterexample`, `collision0_value`, `collision1_value` and `collision` stand at lines 95, 134, 151, 158 and 165 of `Counterexample.lean`, with the content described above. Nothing in `Algebra/JacobianConjecture` concerns iterates, degrees of compositions or dynamics, so the route above starts from nothing formal beyond the map and its baseline identities. The placement of this report beside that Lean and Rocq development confers no formal status on it.

8.3 Consequences that are not claimed

The calculations do not classify unrestricted Keller maps or prove minimal ordinary degree in dimension three. They do not establish an algebraically stable compactification, determine the second dynamical degree, or identify topological entropy with $\log \lambda_1$. The local escape function is not asserted to be a global Green function or a global canonical height. The characteristic-three theorem concerns formal polynomial iteration, not cycle statistics on finite sets. The upper Newton description does not determine all lower faces or all coefficients.

Finally, the existence of a Zariski-dense collection of integral starting points with maximal arithmetic degree does not settle arithmetic-degree questions for all Zariski-dense rational orbits. These distinctions are part of the theorem statements, not post-hoc qualifications to stronger claims.

9 Further research questions and proposed attacks

The following are concrete continuation questions arising from the proved results. They are proposed research problems, not claims that each is a previously named open problem in the literature.

9.1 The second dynamical degree

Determine $\lambda_2(F)$ and $\lambda_2(F_h)$. We now know λ_1 exactly and, in characteristic zero, $\lambda_3 = 3$. The matrix M controls coordinate polynomial degrees, but it is not automatically the action on codimension-two classes. In particular, taking an exterior power of M without constructing a suitable geometric model would not prove a formula for λ_2 . A concrete starting point is to pull back two generic hyperplanes, compute their intersection degrees after removing components at infinity, and search for an exact recurrence that survives saturation.

9.2 A compactification with stable pullback

Construct a projective, toric, or explicitly blown-up model on which the relevant divisor pullback iterates functorially. The wall $a + c = b$ and the two endpoints in [theorem 4.1](#) provide a small amount of combinatorial data from which to start. The challenge is to control centers where the map is indeterminate, not merely to match the Perron eigenvalue. A successful model could explain the recurrence geometrically and support a codimension-two calculation.

9.3 The global escape locus and the complement of the cone

Characterize which complex orbits eventually enter $\mathcal{U}$. The proof gives a genuine open escape region but does not claim that every unbounded orbit enters it. The baseline tropical map has a particularly simple positive-weight chamber structure; the actual logarithmic dynamics can nevertheless encounter cancellation near $xz + 3y = 0$. One approach is to stratify this cancellation region and derive asymptotic transition laws for its successive neighborhoods. A useful first target is an explicit orbit that escapes without eventually approaching the Perron ray, or a proof that no such orbit exists under a stated nondegeneracy condition.

9.4 Arithmetic degree outside the archimedean cone

For rational initial points whose orbits are Zariski dense, decide whether the arithmetic degree always exists and equals the value computed here. An intermediate task is to construct analogous nonarchimedean escape regions and combine their local height contributions. Denominators and bad primes must be tracked: the integral proof avoids them, and therefore does not already give a number-field theorem. A second useful target is an explicit classification of exceptional algebraic subvarieties carrying smaller arithmetic growth.

9.5 Invariant subvarieties and periodic points

Determine invariant curves and surfaces, and compare periodic point counts with degree growth. The rank-two leading matrix and its image plane describe leading exponents; they do not by themselves supply a rational first integral or an invariant hypersurface. Elimination in the ideals generated by $F^n(X) - X$ for small n may reveal patterns, but multiplicities and positive-dimensional components must be separated from isolated periodic points. This question would also clarify when points in the escape region have Zariski-dense individual orbits.

9.6 Dynamical variation under left–right transformations

The source shears give an exact unbounded spectrum within one left–right equivalence class, while the coordinatewise affine changes of [theorem 4.3](#) preserve the baseline value. Classify what happens under coordinate permutations, triangular changes in other directions, and compositions of several shears. A

practical first experiment is to compute the finite list of leading exponent matrices arising from the six coordinate permutations, prove invariant cones when they exist, and locate cases where several tropical chambers recur. This would identify which changes are dynamically harmless and which genuinely alter the asymptotic complexity.

9.7 Beyond the fixed generic-degree-three family

The public weighted-lift audit [6] gives ordinary degree profiles for a different family as its generic degree varies. Its Proposition 8.3 concerns growth across the family parameter, not degrees of iterates. Extend the present no-cancellation method to that family and to the higher-dimensional maps in [5]. The target should be a theorem about an explicitly stated support class or construction, with a verified dominant cone, rather than an extrapolation from a degree profile. This would compare first dynamical degree and generic degree across genuinely different function-field extensions.

9.8 All specializations and higher-order cancellation walls

Characteristic three is completely controlled here because the second coordinate becomes exactly y . For more general coefficient deformations, determine the strata where a leading coefficient vanishes, where two leading terms cancel after composition, or where a chamber changes. The wall identity (4.8) offers a useful test case: its interior support changes in small characteristic even when its endpoints do not. A stronger theory would distinguish loss of interior monomials, loss of a Newton vertex, and change of exponential degree growth.

9.9 The full Newton polytope and coefficient arithmetic

Compute the complete Newton polytopes of F^n , not just their downward completions. The wall initial form supplies exact binomial data on one exposed face, but lower faces may proliferate and encode phenomena invisible to positive weights. One possible route is a finite-state recursion for normal cones outside the positive orthant, accompanied by valuation data for coefficients. A separate arithmetic question is the growth and divisibility of the coefficient vector C_n in (4.7); it is controlled multiplicatively by the same matrix and is accessible to exact integer methods.

9.10 A reusable certified leading-term engine

Implement a checker that accepts a polynomial tuple, a proposed leading exponent matrix, and a rational polyhedral cone, and proves strict dominance plus cone invariance. It should return a kernel-checked degree recurrence when the matrix admits a small annihilating polynomial on the initial degree vector. The baseline and shear family are useful regression tests because one has a two-chamber first step and the other has an unbounded symbolic parameter. Extending the checker to nonmonomial initial forms would make theorem 4.2 a test of more than ordinary monomial-order reasoning.

10 Conclusion

The degree sequence of the repository's three-dimensional map is not merely compatible with $3 + \sqrt{10}$: its entire evolution is forced by a three-by-three exponent matrix and a strict invariant cone. The same proof determines every positive weighted degree and every wall initial form. For polynomial source shears it gives an exact unbounded spectrum while the generic degree stays three. In characteristic three a different exact coordinate identity yields a different, completely explicit phase diagram.

The analytic consequence is equally concrete. The Perron ray is an attracting direction in logarithmic coordinates for a nonempty forward-invariant region, with an exponentially small transverse error after the constant correction. This supplies maximal arithmetic degree for a Zariski-dense set of integral starting points. The remaining global dynamical and arithmetic questions are now separated from the leading-term problem, which has an exact solution for the displayed family.

A Source ledger and priority boundary

A.1 Repository snapshot

The ProveIt reference snapshot is

a866ff9a2cdb9c5f436bb1d82ea5dca59dfee38d.

The main project README, the Jacobian project README, the weighted-classification README, and the arithmetic-fibers README were read through the GitHub connector. The original Lean `Counterexample.lean` was additionally read at this exact commit, including the displayed polynomial tuple, the determinant proof, and the integral-collision declarations. The compiled Lean and Rocq projects were not rerun.

The classification and arithmetic reports are explicitly marked unrefereed in their own status descriptions. Our new proofs need only the displayed definition of F and T_h ; their correctness does not depend on the full classification or on the arithmetic Hasse-principle results in those reports.

A.2 External primary materials

Gao’s arXiv version 2608.00222v1 was reviewed in its HTML representation, especially Section 3.4 and Theorem 3.3. Giannini’s Zenodo landing-page description was accessible and supplied the numerical prediction. Retrieval of its linked PDF failed, and the linked source repository returned not found through the available connector. No claim is made about unseen sections of that PDF.

The external audit [6] was also inspected, in particular Proposition 8.3. Its formula $(5d - 3, 5d - 4, 4)$ concerns component degrees as the degree d of a construction polynomial varies. It is not an iterate recurrence. The retrieved file had content blob identifier a6950580cd582458d1f05ccf259af33e833f9fce.

Targeted searches for the map together with dynamical-degree and degree-growth terminology did not establish whether any unreviewed or later source already proves one of the results here. Consequently this is an independently derived research article, not an exhaustive priority determination. In particular, “the suggested value is proved” is the supported claim; “the first proof of a long-standing open conjecture” is not.

B Reproduction and package contents

The article is self-contained LaTeX with an embedded bibliography. A standard TeX installation providing `newtxtext`, `newtxmath`, `tcolorbox`, and the usual mathematics packages can build it with two runs of `pdflatex` (a third run is harmless). The companion verification uses SymPy 1.14.0:

```
python -m pip install -r requirements.txt
python code/verify_results.py
pdflatex -interaction=nonstopmode -halt-on-error article.tex
pdflatex -interaction=nonstopmode -halt-on-error article.tex
```

A **Makefile** provides the equivalent targets. Do not run the Python verifier with optimization flags that disable assertions; its exact checks intentionally use assertions.

The archive includes the source and compiled PDF, the exact verification script and its recorded outputs, a compact degree calculator, and the files `README.md`, `SOURCES.md`, `STATUS.md`, and `SHA256SUMS`. The source ledger identifies inherited facts, the status file states the proof boundaries, and the checksums identify the delivered artifacts. Font files are not included. Generated TeX auxiliary files and page-preview images are not part of the delivery.

[write] The two paragraphs above describe the delivered package. In ProveIt the layout differs, and some of the commands above must not be run as printed.

- In ProveIt the requirements file is `data/requirements.txt`, and the Makefile is `code/Makefile`. The Makefile's targets name `code/verify_results.py` and `article.tex` relative to the package root, so they fail when run from `code/`.
- `code/verify_results.py` writes its record, by default, to `data/verification.json`: run as printed it overwrites the shipped record (on Windows with CRLF line endings). Run it on a copy, or pass `--output` with a path outside the report. Its recorded console output `data/verification.txt` ends with the delivery path `/mnt/data/ProveIt_Keller_Dynamics/data/verification.json`.
- The delivered `README.md`, the compiled PDF and `SHA256SUMS` are not shipped: the `README` is replaced by the report's own, `article.pdf` is a build of the present text, and the checksum manifest was verified (13 of 13 entries) and retired. The `article_sha256` recorded in `data/build_review.json` is the hash of the delivered PDF, not of `article.tex`, and that record describes the delivered 23-page build.
- `data/calculator_checks.json` (930 degree vectors) is produced by no shipped script; it is kept as delivered.

C Provenance of this report

[write] This report is built from *one* manuscript: manuscript 06 of batch 70 of ProveIt's incoming reports, the archive `ProveIt_Keller_Dynamics` (inner directory of the same name, main file `article.tex`, a 23-page A4 PDF dated 30 September 2026, PDF metadata author "ChatGPT; prepared for Vladimir Reshetnikov"), pinned to ProveIt commit `a866ff9a2cdb9c5f436bb1d82ea5dca59dfee38d`. The archive arrived in commit `1b3960d8a` and was placed in `51c6943bf` as a new report of the category `jacobian-conjecture`. Between the pin and the placement commit nothing under `Algebra/JacobianConjecture` or under the category's other reports changed. Nothing was merged and nothing was selected away: every result, proof, example, table, question and limitation of the manuscript is printed, and the manuscript contributed everything except the text marked [write].

The one choice made at placement was where the manuscript goes. It studies the map of the formal project and the shears of [2], but answers no question that [2] or [3] asks, so it became a report of its own rather than a Part III of [2].

The editorial changes are these.

- (1) Every label carries the prefix `kmd:`. The manuscript's 87 labels are kept, unchanged after the prefix, and every reference to them was updated; one label was added, `kmd:app:provenance:88` in all. The manuscript's macros are kept as delivered.
- (2) Notes marked [write] were added on the title page, after (1.2) (notation against the rest of ProveIt and the formalized declarations), after the remark following [theorem 2.2](#) (prior sources of the cubic

and of generic degree three), after the table following [theorem 5.2](#) (the case $n = 1$ is Corollary 6.4 of [2]), after [theorem 5.3](#), at the end of the formalization route in [section 8](#), and in Appendix B (shipped layout).

- (3) The title page no longer creates a PDF page anchor, so the title page and the first numbered page no longer share one.

No mathematical statement, proof or number of the manuscript was changed, and the external sources [4, 5, 6] were not re-checked at the write.

References

- [1] Vladimir Reshetnikov and contributors. *ProveIt: Jacobian-conjecture development*. Repository documentation and Lean source at commit [a866ff9a2cdb9c5f436bb1d82ea5dca59dfee38d](#). [Repository directory at the pinned commit](#).
- [2] ProveIt research-report collection. *All-Degree Rigidity of a Weighted Keller Class*, Parts I–II. AI-assisted, unrefereed reports dated 24 and 29 September 2026; README and status ledger reviewed. [Repository directory at the pinned commit](#).
- [3] ProveIt research-report collection. *Arithmetic local–global fibers*, Parts I–III. AI-assisted, unrefereed reports; README and notation ledger reviewed. [Repository directory at the pinned commit](#).
- [4] Liam Giannini. *The Alpöge–Fable counterexample to the Jacobian Conjecture: verification, geometry, dynamics, and an equivariant program for the plane case*. Zenodo, version 1.1, 20 July 2026. DOI: [10.5281/zenodo.21461572](#). Public record description reviewed; full PDF not successfully retrieved.
- [5] Shuhong Gao. *Counterexamples to the Jacobian conjecture in dimensions greater than two*. arXiv:2608.00222v1, 31 July 2026, Section 3.4 and Theorem 3.3. <https://arxiv.org/html/2608.00222v1>.
- [6] shadybrook/jacobian-counterexample-audit contributors. *An Exact Audit of an Announced Three-Dimensional Keller Map*. AI-assisted working note, stated revision 21 July 2026; Proposition 8.3 and adjacent discussion reviewed. [Public working-note source](#).